HYPERION AI

SYSTEM DESIGN DOCUMENT

AI Pipeline System

Async Python pipelines at scale - FastAPI gateway, Celery workers, dedicated DLQ, structured logging

PREPARED FOR

Teams running many async Python tools in parallel

Vitalijus Alsauskas

Software and AI Engineer

Hyperion AI - Production AI engineering for founders and agencies

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Async Python Pipelines at Scale — System Design

Prepared for	Teams running 19+ async Python tools in parallel
Prepared by	Vitalijus Alsauskas · Hyperion AI
Scope	FastAPI gateway, Celery + Redis orchestration, isolated workers per tool, dedicated DLQ, structured logging with request IDs

1. Problem Framing

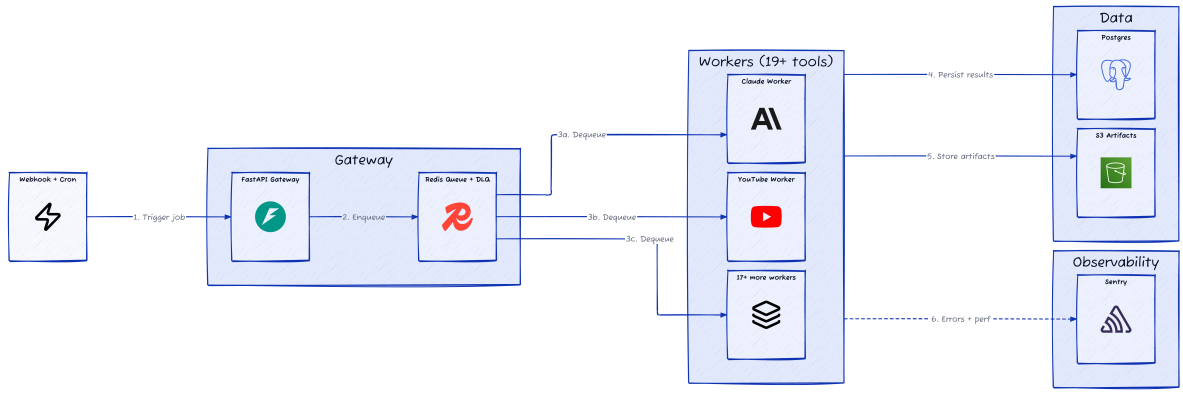
Async Python at 2-3 tools feels like a solved problem. At 10+ it falls over.

Common failure modes once a pipeline fleet grows:

- **One bad worker starves the rest** — the video-render queue backs up, takes down the LLM queue with it
- **Silent retry failures** — retries look like success, exhausted-retry jobs just vanish
- **Scattered logs across 19 streams** — 3am debugging turns into grep across a dozen services
- **No one knows who called what** — no trace of which trigger spawned which worker + external API calls

This design shows the pattern that **holds at scale**: FastAPI front-door, Celery workers isolated per tool, a dedicated DLQ that never silently loses jobs, and `request_id` propagation that makes one grep enough to reconstruct any job's path.

2. System Diagram



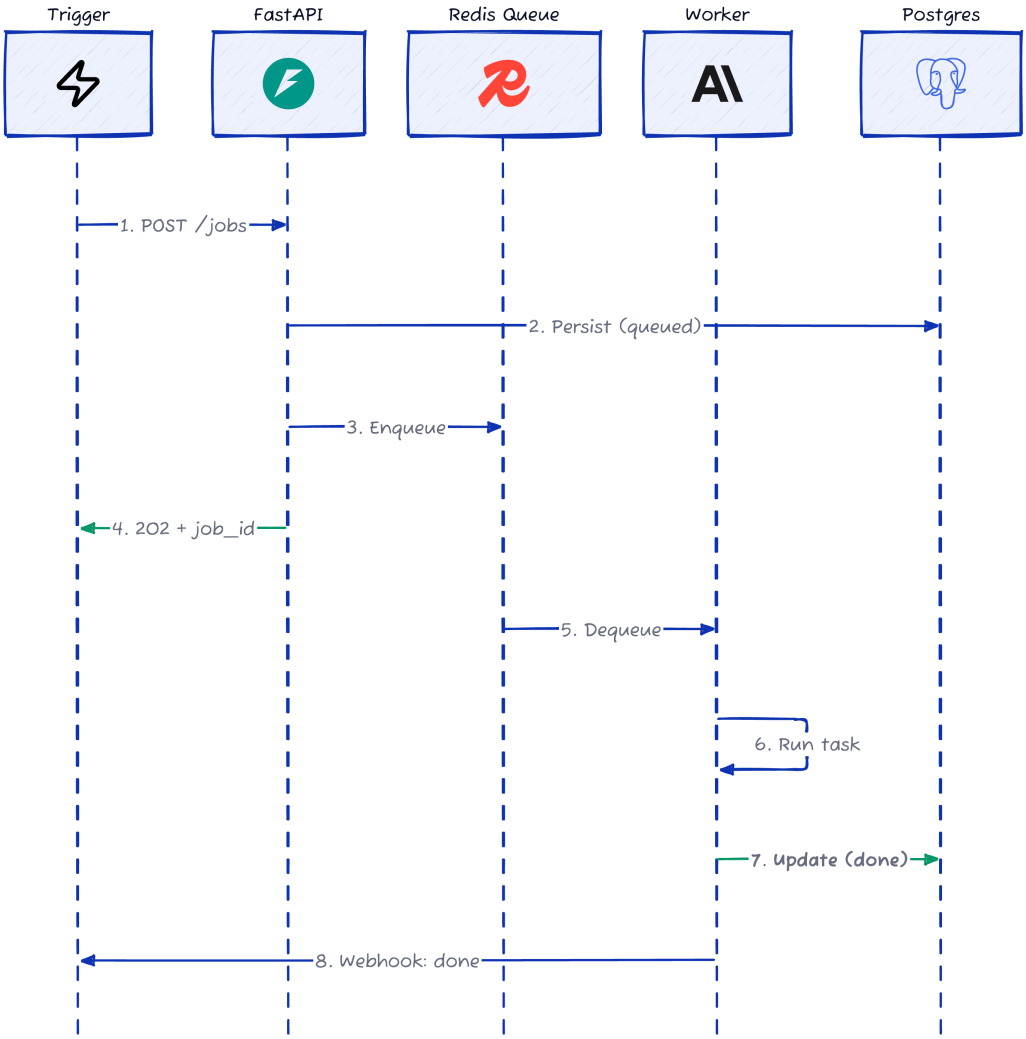
Every pipeline tool runs as **its own isolated Celery worker**. A video-render failure never blocks Claude extraction. Every exhausted-retry job lands in a **dedicated DLQ** — never silently lost.

Source: `docs/architecture.d2`

3. Tool Choices

LAYER	PICK	WHY THIS	ALTERNATIVES CONSIDERED
API Gateway	FastAPI + Pydantic	Async-native, OpenAPI auto, strong typing from day one	Flask (no async), Django (overkill), aiohttp (weaker DX)
Task Queue	Celery + Redis	Mature, battle-tested, fine-grained retry + DLQ control	RQ (simpler, weaker scheduling), Dramatiq (smaller ecosystem), Arq (young)
Broker	Redis	Fast, cheap, doubles as cache	RabbitMQ (more ops), SQS (cloud lock-in)
Scheduler	Celery Beat	Co-located with Celery	APScheduler (extra service), cron (no retry visibility)
Workers	Celery, one class per tool	Isolation, per-tool concurrency tuning	Monolithic worker (one bad job starves rest)
Observability	Structlog + Sentry + Flower	JSON logs, exception context, queue visibility	ELK (overkill), DataDog (expensive)
DB	Postgres	Job state + results, strong consistency	Mongo (worse for relational graphs), Redis-only (no durability)
Object Storage	S3	Standard, cheap	Local disk (doesn't scale)
LLM	Claude 3.5 Sonnet	Strong structured output, zero-retention	GPT-4o (close, retention less favorable), local Llama (unreliable at scale)

4. Data Flow



Happy path

1. **Trigger** fires (cron, webhook, or manual CLI)
2. **API** validates and persists job row with `status=queued`
3. Job ID **enqueued** to Redis broker
4. **Worker** for the appropriate tool dequeues
5. Worker calls external APIs (Claude, YouTube, ffmpeg, etc.)
6. Artifacts land in **S3**, small results in **Postgres**
7. Job row updated to `status=done`
8. Outbound **webhook** fires to the downstream system

Error paths

FAILURE TYPE	HANDLING
Retryable (rate limit, network timeout, 5xx)	Caught, logged to Sentry with context, requeued with exponential backoff up to retry budget (default 3)
Non-retryable (malformed input, auth failure)	Marked failed immediately, logged, no retry
Exhausted retries	Moved to **DLQ**, `status=failed` with error context, Sentry critical alert, failure webhook fires
Worker crash mid-job	Job ack only after successful commit — crashed worker means job returns to queue automatically
DB write failure	Transaction rollback, `status=failed-transient`, retried

5. Component Breakdown

COMPONENT	STACK	RESPONSIBILITY
API Gateway	FastAPI + Pydantic + Structlog	Public job submission + status, auth middleware, audit log
Celery Beat	Celery scheduler	Cron-triggered job submission
Worker Pool	One Python module per tool under `workers/`	Each worker declares concurrency + retry + timeout in code
DLQ Monitor	Lightweight service	Reads DLQ, fires Sentry alerts, exposes `replay` endpoint
Observability Stack	Structlog + Sentry SDK + Flower	JSON logs, async-aware exception capture, queue visibility
Data Layer	SQLAlchemy + Alembic + boto3 + redis-py	Postgres schema, S3 access, broker + cache

6. Implementation Phases

PHASE	WEEKS	SHIPS	KEY MILESTONES
1. Foundation	1-2	Skeleton with one pipeline tool end-to-end	`/jobs` endpoint, Celery + Redis + Beat via docker-compose, one worker class, Structlog + Sentry + Flower, DLQ monitor, integration tests
2. Pipeline Coverage	3-4	All 19+ tools migrated into isolated Celery workers	Per-tool retry/timeout policies, load test at 3× peak, per-worker runbook stubs
3. Production Polish	5-6	Observability refined, DLQ replay flow, prod deploy	Request ID propagation across all workers, DLQ replay CLI, full runbook, `/replay-dlq` Claude Code command, GitHub Actions deploy

7. Risks & Mitigations

#	RISK	MITIGATION
1	**Malformed input blocks the whole queue**	Worker isolation — each tool has its own queue + pool. Hard timeouts on every task prevent hung APIs from pinning workers.
2	**Retries mask silent failures** — jobs vanish	Dedicated DLQ. Every exhausted-retry job lands there, fires Sentry critical, marks `status=failed` with full context, fires failure webhook. Never silently lost.
3	**3am debugging across 19 log streams**	Structlog + `request_id` attached at ingress, propagated everywhere. One `grep request_id=abc123` surfaces the full job history.
4	**Claude API rate limits during peak**	Per-worker concurrency caps + exponential backoff on 429 + Claude-specific rate limiter. Haiku fallback under extreme load.
5	**Autoscaling misconfigured** — jobs pile up at night	Queue-depth alerts (warn 100, critical 500). Autoscaling tied to queue depth not CPU. Flower accessible from phone.

8. What I Need From You

To move into Phase 1, a few inputs from your side:

- **Read access to the existing codebase** (or a representative sample) — so I can read before proposing changes
- **List of the 19+ tools** — one-liner per tool on what each does
- **Current deployment target** — Hetzner VM? Container Apps? k8s? bare metal?
- **Ranked pain points** — is 3am debugging the worst, or rate limits, or onboarding new pipelines?

- **30-minute call** — to walk through this doc against your actual setup and tighten scope before code ships

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This document is yours to keep regardless of whether we end up working together. The architecture and tool choices here should save any developer who inherits this work a week of deliberation.